

Guanine Exchange Factors Show Existence of Transient Pockets on Rac1 Surface

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Supplementary Information

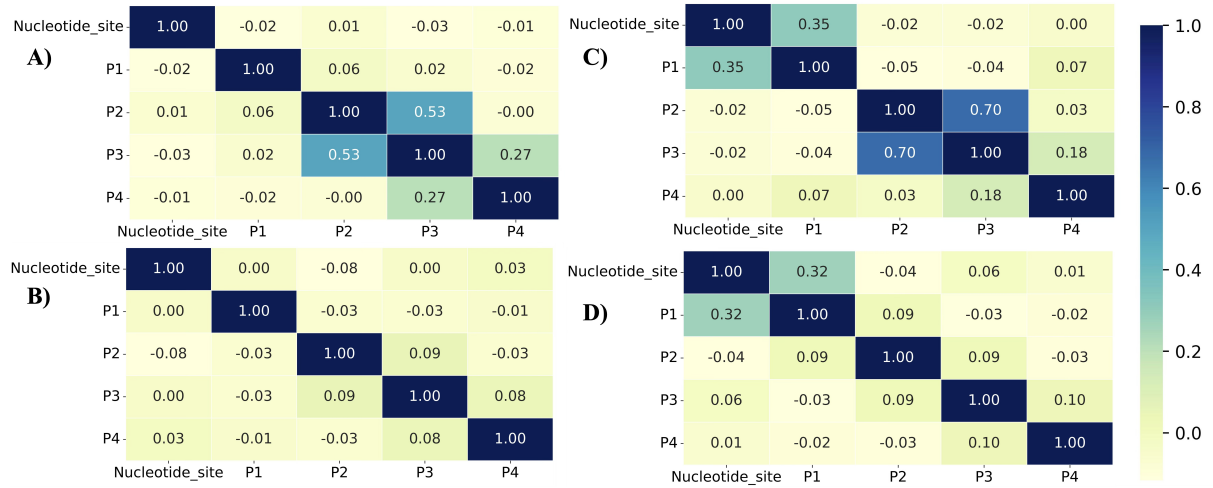


Figure 1: Pearson Correlation for the volume fluctuations among different pockets and the effect of nucleotide binding. **A)** GDP-bound Rac1; **B)** GTP-bound Rac1; **C)** Nucleotide-free state created by deleting GDP from Rac1; **D)** Nucleotide-free state created by deleting GTP from Rac1.

Table 1: Rac1 residues lining Site P1 with residuewise energy contribution to GEF binding

Residue	PREX1	TIAM1	VAV1	DOCK1	Comment
T17	Y	Z	X1	Y1	
L20	Y	Z	X1	Y1	
T35	Y	Z	X1	Y1	
V36	Y	Z	X1	Y1	
F37	Y	Z	X1	Y1	
D38	Y	Z	X1	Y1	
N39	Y	Z	X1	Y1	
L55	Y	Z	X1	Y1	
W56	Y	Z	X1	Y1	
D57	Y	Z	X1	Y1	
T58	Y	Z	X1	Y1	
A59	Y	Z	X1	Y1	
G60	Y	Z	X1	Y1	
Q61	Y	Z	X1	Y1	
S71	Y	Z	X1	Y1	

Table 2: Rac1 residues lining Site P2 with residuewise energy contribution to GEF binding

Residue	PREX1	TIAM1	VAV1	DOCK1	Comment
T17	Y	Z	X1	Y1	
L20	Y	Z	X1	Y1	
T35	Y	Z	X1	Y1	
V36	Y	Z	X1	Y1	
F37	Y	Z	X1	Y1	
D38	Y	Z	X1	Y1	
N39	Y	Z	X1	Y1	
L55	Y	Z	X1	Y1	
W56	Y	Z	X1	Y1	
D57	Y	Z	X1	Y1	
T58	Y	Z	X1	Y1	
A59	Y	Z	X1	Y1	
G60	Y	Z	X1	Y1	
Q61	Y	Z	X1	Y1	
S71	Y	Z	X1	Y1	

Table 3: Rac1 residues lining Site P3 with residuewise energy contribution to GEF binding

Residue	PREX1	TIAM1	VAV1	DOCK1	Comment
T17	Y	Z	X1	Y1	
L20	Y	Z	X1	Y1	
T35	Y	Z	X1	Y1	
V36	Y	Z	X1	Y1	
F37	Y	Z	X1	Y1	
D38	Y	Z	X1	Y1	
N39	Y	Z	X1	Y1	
L55	Y	Z	X1	Y1	
W56	Y	Z	X1	Y1	
D57	Y	Z	X1	Y1	
T58	Y	Z	X1	Y1	
A59	Y	Z	X1	Y1	
G60	Y	Z	X1	Y1	
Q61	Y	Z	X1	Y1	
S71	Y	Z	X1	Y1	

Table 4: Rac1 residues lining Site P4 with residuewise energy contribution to GEF binding

Residue	PREX1	TIAM1	VAV1	DOCK1	Comment
T17	Y	Z	X1	Y1	
L20	Y	Z	X1	Y1	
T35	Y	Z	X1	Y1	
V36	Y	Z	X1	Y1	
F37	Y	Z	X1	Y1	
D38	Y	Z	X1	Y1	
N39	Y	Z	X1	Y1	
L55	Y	Z	X1	Y1	
W56	Y	Z	X1	Y1	
D57	Y	Z	X1	Y1	
T58	Y	Z	X1	Y1	
A59	Y	Z	X1	Y1	
G60	Y	Z	X1	Y1	
Q61	Y	Z	X1	Y1	
S71	Y	Z	X1	Y1	

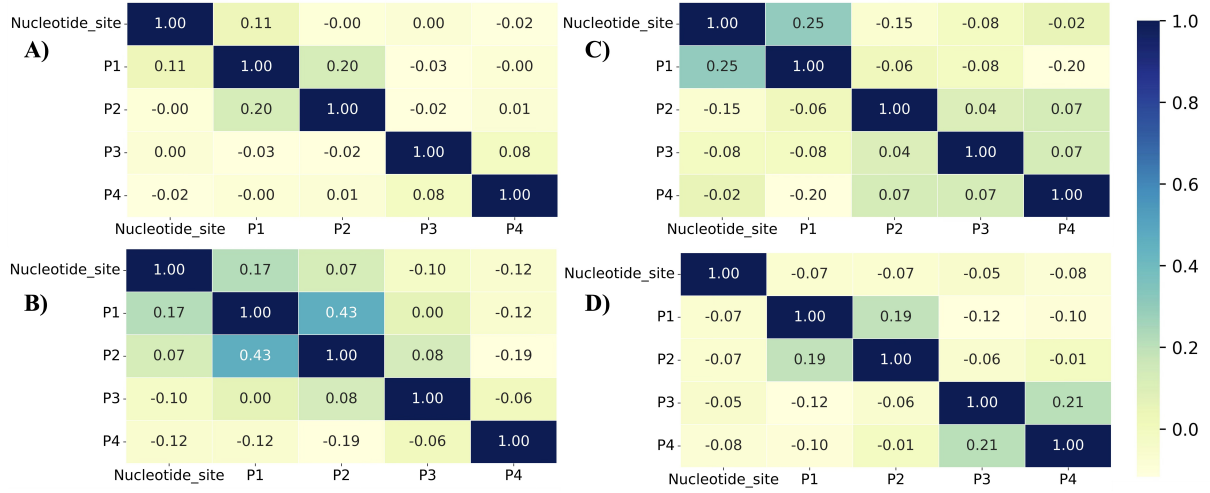


Figure 2: Pearson Correlation for the volume fluctuations among different pockets and the effect of GEF binding. **A)** PREX1-bound Rac1; **B)** TIAM1-bound Rac1; **C)** VAV1-bound Rac1; **D)** DOCK2-bound Rac1. Rac1 in all GEF-bound systems is in the nucleotide-free state.